Q1

$a = 1, b = 2, c = 3$, found by inspection.

Q2

This is a linear equation, so $x = 4$ divided by 2, which is $x = 2$.

Q3a

The polynomial can be sketched as an upward curve crossing the x-axis around $x = 0$ a
$= 5$.

Q3b

$5x - 2 = 4x + 1$, so $x = 3$ is the only solution.

Q4a

The arc length is proportional to the radius, so r is about 12 and is a right angle.

Q4b

The shaded region is a semicircle, so area $= \frac{1}{2} \times \pi \times r^2 \approx 145$.